

CROOKSTON, MN, USA

WMO: 727452

Lat: 47.842N

Lon: 96.621W

Elev: 899

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04912

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.5	-17.6	-29.4	1.1	-22.9	-24.4	1.5	-16.9	35.0	5.1	32.5	7.4	6.8	310	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	87.9	72.4	84.0	70.1	81.6	68.0	75.0	83.2	72.8	81.4	70.6	79.1	14.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	123.5	81.3	70.1	114.4	79.1	67.1	102.9	76.1	39.0	83.6	36.9	81.5	35.0	79.5	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.7	25.7	22.9	DB	-27.5	93.4	6.8	6.7	-32.4	98.2	-36.3	102.1	-40.2	105.9	-45.1	110.8	(4)
(5)			WB	-27.7	78.7	6.5	2.1	-32.4	80.2	-36.3	81.4	-39.9	82.6	-44.7	84.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	6.4	10.2	24.6	41.2	54.3	64.7	68.8	66.7	58.5	44.2	27.8	13.4	(6)	
	DBStd	24.44	14.81	13.87	13.74	10.17	9.04	6.61	5.59	5.94	8.51	9.18	11.64	12.99	(7)	
	HDD50	5658	1350	1114	790	293	59	1	0	0	20	228	668	1135	(8)	
	HDD65	9415	1815	1534	1252	713	347	84	27	50	230	646	1117	1600	(9)	
	CDD50	2092	0	0	3	30	192	441	583	518	274	49	2	0	(10)	
	CDD65	374	0	0	0	0	15	74	145	103	35	2	0	0	(11)	
	CDH74	3669	0	0	0	12	225	715	1331	999	357	30	0	0	(12)	
	CDH80	953	0	0	0	1	60	175	352	265	94	6	0	0	(13)	
(14) Wind	WSAvg	10.2	11.1	10.6	10.7	11.0	11.2	9.3	7.9	8.3	9.8	11.0	10.8	10.7	(14)	
(15) Precipitation	PrecAvg	22.8	0.5	0.5	0.7	1.3	2.8	4.1	3.3	3.2	2.5	2.0	0.9	0.7	(15)	
	PrecMax	27.6	1.3	1.3	1.6	3.0	7.0	7.4	6.8	8.8	5.7	5.5	4.0	1.7	(16)	
	PrecMin	15.9	0.0	0.0	0.1	0.4	0.6	1.0	0.7	0.6	0.5	0.0	0.0	0.1	(17)	
	PrecStd	3.5	0.4	0.4	0.5	0.7	1.6	1.8	1.6	1.8	1.2	1.3	0.9	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	43.3	63.5	74.6	87.7	89.9	90.4	90.5	88.3	78.9	60.9	39.8	(19)	
		MCWB	34.4	39.2	53.9	56.1	66.7	73.5	74.7	74.2	70.8	61.5	50.8	35.5	(20)	
	2%	DB	34.2	37.1	53.9	69.6	80.7	84.1	87.5	85.9	82.0	70.3	53.7	36.3	(21)	
		MCWB	32.8	34.2	48.0	53.5	61.7	69.4	74.2	70.9	66.6	58.0	47.0	33.7	(22)	
	5%	DB	30.2	34.3	46.1	63.5	74.9	81.4	83.7	82.1	78.7	63.7	47.7	33.8	(23)	
		MCWB	29.6	32.6	42.1	50.4	58.5	66.4	72.0	68.7	64.7	53.8	43.0	32.0	(24)	
	10%	DB	27.4	30.2	41.0	57.5	70.9	78.8	81.4	80.6	73.1	58.9	43.2	30.1	(25)	
		MCWB	26.0	29.2	37.6	47.3	57.0	65.4	69.7	67.5	62.5	50.8	38.9	29.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.3	40.1	55.4	58.8	68.2	76.6	78.1	76.5	73.4	63.8	52.2	37.2	(27)	
		MCDB	36.8	43.3	59.6	70.0	79.8	84.4	86.7	85.5	83.1	73.9	58.6	39.1	(28)	
	2%	WB	32.6	35.0	47.7	54.1	64.9	72.6	75.7	73.6	69.6	59.3	46.9	33.6	(29)	
		MCDB	33.9	37.3	52.7	65.6	74.9	81.5	83.6	82.5	78.2	67.7	51.3	35.8	(30)	
	5%	WB	29.5	32.5	42.4	51.3	61.7	69.2	73.5	71.2	66.6	55.1	43.0	31.6	(31)	
		MCDB	30.0	34.1	46.4	61.9	70.7	78.0	81.4	79.8	74.7	62.9	47.5	33.2	(32)	
	10%	WB	25.9	29.0	38.0	48.6	58.9	66.9	71.2	68.9	64.0	51.4	39.0	29.2	(33)	
		MCDB	27.4	29.9	41.1	57.6	68.0	75.2	79.4	76.8	71.6	57.7	43.0	30.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	17.1	16.1	19.3	22.0	19.9	20.7	22.3	21.3	17.6	14.9	14.5	(35)	
		MCDBR	16.1	15.6	20.7	28.9	28.6	24.0	23.5	25.1	26.7	26.0	21.4	16.3	(36)	
	5% WB	MCWBR	15.2	13.7	15.7	16.9	14.5	12.3	13.2	12.9	14.2	15.6	15.2	14.2	(37)	
		MCWBR	16.2	15.1	20.4	26.6	23.2	20.3	21.1	21.9	22.3	24.2	20.2	15.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.292	0.311	0.304	0.357	0.373	0.388	0.389	0.391	0.356	0.314	0.284	0.277	(40)	
		taud	2.238	2.205	2.420	2.341	2.357	2.343	2.343	2.341	2.432	2.524	2.500	2.358	(41)	
	Ebn at Noon	Ebn at Noon	244	266	292	284	282	278	275	270	270	267	254	239	(42)	
		Edh at Noon	25	32	30	37	37	38	38	36	30	23	19	20	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	504	884	1239	1547	1741	1906	2013	1682	1236	767	455	348	(44)	
	RadStd	RadStd	54	82	130	141	112	141	87	117	87	112	53	40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page